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Optimization of safety equipment outages improves safety

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Abstract

Testing and maintenance activities of safety equipment in nuclear power plants are an important potential for risk and cost reduction. An optimization method is presented based on the simulated annealing algorithm. The method determines the optimal schedule of safety equipment outages due to testing and maintenance based on minimization of selected risk measure. The mean value of the selected time dependent risk measure represents the objective function of the optimization. The time dependent function of the selected risk measure is obtained from probabilistic safety assessment, i.e. the fault tree analysis at the system level and the fault tree/event tree analysis at the plant level, both extended with inclusion of time requirements. Results of several examples showed that it is possible to reduce risk by application of the proposed method. Because of large uncertainties in the probabilistic safety assessment, the most important result of the method may not be a selection of the most suitable schedule of safety equipment outages among those, which results in similarly low risk. But, it may be a prevention of such schedules of safety equipment outages, which result in high risk. Such finding increases the importance of evaluation speed versus the requirement of getting always the global optimum no matter if it is only slightly better that certain local one. © 2002 Elsevier Science Ltd. All rights reserved.

Keywords: Safety; Reliability; Probabilistic safety assessment; Risk; Optimization

1. Introduction

Safety is one of the most important features of the complex facilities, such as nuclear power plants. High level of safety is achieved by the implementation of numerous safety standards and safety analyses [1]. Assessment of safety and risk reduction is mostly connected with risk based methods, e.g. probabilistic safety assessment [2]. Testing and maintenance activities of safety equipment in nuclear power plants are an important potential for risk [3]. This potential has been realized by many researchers [4], and substantial amount of work has been performed, including:

- optimization of replacement times, where Schabe developed an approach for calculation of optimal replacement time of maintainable components in a complex system based on minimal life cycle costs [5];
- optimization of inspection strategy for randomly failing equipment by Chelby et al. [6];
- optimization of test intervals [7–10], where Martorell et al. [7], Čepin et al. [8] and Vaurio [9] developed several methods for optimization of test intervals based on risk or costs;
- optimization of test strategies [11–13], where Hokstad et al. [11] developed a reliability model for optimization of test schemes for fire and gas detectors, which is based on minimal system unavailability.

As a consequence of the results from such risk based methods, test intervals and allowed outage times of safety equipment were changed in some nuclear power plants technical specifications [14–16].

Less has been done on optimization of time arrangement of testing and maintenance activities, although the issue was addressed by an example of 10 component high pressure injection system, where dynamic programming approach was used to minimize the costs of maintenance by Harunuzzaman et al. [17].

Common to those ideas and their implementations is a fact that the problems are becoming larger and larger, because the new methods and better computers in present time allow modeling of processes and properties, which were not considered in the past. In that sense, the simulated annealing methods and genetic algorithms have been found extremely useful for optimization problems of large scale with desired global extremum hidden among many local extrema [7,18,19].

The purpose of this study is to propose a methodology to reduce the risk contribution of testing and maintenance

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activities of safety equipment in a nuclear power plant by optimization of time scheduling of safety equipment outages due testing and maintenance. Such optimization is an important issue due to lack of methods and their applications on scheduling the testing and maintenance of safety equipment based on minimal risk.

This paper presents an optimization method, which integrates simulated annealing with probabilistic safety assessment to minimize risk of testing and maintenance activities of safety equipment in a nuclear power plant by their optimal scheduling. Probabilistic safety assessment and simulated annealing are described in Section 2. The optimization algorithm, which calculates the minimal risk dependent on the time arrangement of safety equipment outages due to testing and maintenance, is defined in Section 3. Performance tests, which serve for verification and assessment of the algorithm, are presented in Section 4. The results of optimization are discussed in Section 5. Findings and implications of the study are summarized in Section 6.

2. Methods

The following prerequisite for the optimization of safety equipment outages based on minimal risk was used:

> Optimization method should use the models and results of probabilistic safety assessment, which is a comprehensive source of safety information.

Such an approach necessitates the upgrade of probabilistic safety assessment models with time requirements [20] and leads to the extended use of their results in accordance with current trend of risk-informed decision-making [21].

Current approach to probabilistic safety assessment is summarized in Section 2.1. Simulated annealing as the mathematical optimization method is summarized in Section 2.2.

2.1. Probabilistic safety assessment

Probabilistic safety assessment is one of the standardized ways of assessing safety in nuclear power plants [2,8]. Its primary methods: fault tree and event tree analysis, include a wide range of applications [22].

Fault tree is a tool to identify and assess all combinations of undesired events in the context of system operation and its environment that can lead to the undesired state of a system [23]. Undesired state of the system is represented by a top event. Logical gates connect the basic events to the top event. Basic events are the ultimate parts of the fault tree, which represent undesired events, such as component failures, missed actuation signals, human errors, contributions of testing and maintenance activities and common cause contributions.

Qualitative fault tree analysis identifies the minimal cut sets, which are the combinations of the smallest number of component faults that may cause the system fault. In other words: the minimal cut sets are combinations of the smallest number of basic events, which, if occur simultaneously, may lead to the top event.

Quantitative fault tree analysis includes calculation of the system unavailability (calculated as the fault tree top event probability) which is one of the main risk measures at the system and component level and which is based on probability of failure of safety system components [2,23,24].

Event tree is a tool to identify and assess possible scenarios (i.e. accident sequences) of safety systems functions responses (i.e. systems successes or system faults, which are further analyzed with the fault trees) to the initiating event [2]. The initiating event is an event which may lead to the accident consequences. Safety system functions are the means to prevent the accident or to mitigate its consequences. Plant damage states are the end states of the scenarios.

In general, two main approaches for the event tree analysis exist: (a) fault tree event tree linking approach (known also as large fault tree small event tree approach) and (b) event tree with boundary conditions approach (known also as small fault tree large event tree approach) [2,24]. The first approach is used in this study. The qualitative results of the event tree analysis include minimal cut sets for accident sequences. The quantitative results include accident sequences frequencies, which are combined together through all respective scenarios and through all the event trees developed for the plant level analysis.

2.1.1. Probabilistic modeling of components

Probabilistic models of components represent the standpoint for probabilistic modeling of systems. There exist a number of probabilistic models for components either constant in time or time dependent, such as probability per demand model and constant failure rate model [2,8,25]. Input data to probabilistic models differ from model to model regarding the function and operation of the component under consideration. Input data may include a number of parameters, such as failure rate for the component, probability of failure per demand, repair time (restore time), test interval, test duration time and test placement time [2,8,23–25].

The prerequisites for optimization of time scheduling of safety equipment outages due testing and maintenance are the time dependent probabilistic models for modeling of the system components (e.g. represented by basic events in the fault trees) [20].

The universal time dependent probabilistic model for modeling of component failure probabilities was selected [8], although any other time dependent probabilistic

model could be used instead of the selected one:

$$
\begin{aligned}
&0 < t < (T_p - T_t f_o); \qquad Q_{Bj}(t) = q + 1 - e^{-\lambda t},\\
&(T_p - T_t f_o) < t < T_p; \qquad Q_{Bj}(t) = 1,\\
&0 < ((t - T_p) \bmod T_i) < (T_i - T_t f_o - T_r);\\
&Q_{Bj}(t) = q + 1 - e^{-\lambda((t-T_p) \bmod T_i)},\\
&(T_i - T_t f_o - T_r) < ((t - T_p) \bmod T_i) < (T_i - T_t f_o);\\
&Q_{Bj}(t) = q + 1 - e^{-\lambda T_i} + (e^{-\lambda T_i} - q)(q + 1\\
&\qquad - e^{-\lambda(((t-T_p) \bmod T_i) - T_i)}),\\
&(T_i - T_t f_o) < ((t - T_p) \bmod T_i) < T_i; \qquad Q_{Bj}(t) = 1
\end{aligned}
\tag{1}
$$

where $Q_{Bj}(t)$ is the time dependent failure probability for component Bj, λ the failure rate, T_i the test interval, T_r the mean time to repair (restore), T_p the test placement time, T_t the test duration time, T the time, q the probability per demand, f_o the test override factor, and mod is the function: dividing reminder.

Outcomes of Eq. (1) are the following:

- By putting all times (T_i, T_p, T_r, T_t) to 0 only probability for demand defines the unavailability of a component. In that case the equation calculates unavailability of a standby component functioning on demand.
- By introducing the data for times T_i and T_t, the equation corresponds to the probabilistic model for periodically tested standby components.
- By introducing T_r the equation is expanded with the unavailability contribution of mean time to repair.
- By introducing T_p, the evaluation of different testing strategies for redundant components and risk impact of tests scheduling becomes possible.
- By introducing f_o it is possible to model that component is unavailable only a fraction of test duration time.

2.1.2. Probabilistic modeling of systems

The modeling of safety systems follows the fault tree analysis [2,23,24] and its extension with time, i.e. dynamic fault tree analysis [20], where the fault tree top event probabilities are assessed using the following expressions [20]:

$$
\begin{aligned}
Q_{GD}(t) = &\sum_{i=1}^{n} Q_{MCS_i(H_{1t},\ldots,H_{st},\ldots,H_{St})}\\
&- \sum_{i<j} Q_{MCS_i(H_{1t},\ldots,H_{st},\ldots,H_{St}) \cap MCS_j(H_{1t},\ldots,H_{st},\ldots,H_{St})}\\
&+ \sum_{i<j<k} Q_{MCS_i(H_{1t},\ldots,H_{st},\ldots,H_{St}) \cap MCS_j(H_{1t},\ldots,H_{st},\ldots,H_{St}) \cap MCS_k(H_{1t},\ldots,H_{st},\ldots,H_{St})}\\
&- \cdots + (-1)^{n-1} Q_{\bigcap_{i=1}^{n} MCS_i(H_{1t},\ldots,H_{st},\ldots,H_{St})}
\end{aligned}
\tag{2}
$$

from which only the first three factors are considered in this study (at the probabilistic modeling of the complete plant only the first factor is usually considered) and where (basic events in the fault tree are assumed mutually independent)

$$
Q_{MCS_i}(H_{1t}, H_{2t}, \ldots, H_{st}, \ldots, H_{St}) = \prod_{j=1}^{m} Q_{Bj}(t) \tag{3}
$$

where $Q_{GD}(t)$ is the top event probability at time t, MCS_i the minimal cut set i, $Q_{MCS_i}(H_{1t}, H_{2t}, \ldots, H_{st}, \ldots, H_{St})$ the probability of ith minimal cut set at time t, $Q_{Bj}(t)$ the probability of occurrence of basic event Bj at time t, n the number of minimal cut sets, m the number of basic events in minimal cut set i, and H_{st} is the value of house event s at time t.

2.1.3. Probabilistic modeling of the complete plant

Probabilistic modeling of the complete plant represents combining together all event trees, which combine all fault trees. The accident sequences in the respective event tree are expressed as

$$
F_{ASj}(t) = F_{IE} \sum_{i=1}^{n} Q_{MCS_i}(H_{1t}, H_{2t}, \ldots, H_{st}, \ldots, H_{St}) \tag{4}
$$

where $F_{ASj}(t)$ is the frequency of accident sequence j at time t, F_{IE} the initiating event frequency, $Q_{MCS_i}(H_{1t}, H_{2t}, \ldots, H_{st}, \ldots, H_{St})$ the probability of ith minimal cut set at time t, and n is the number of minimal cut sets.

Plant damage state frequencies combine together the respective accident sequences frequencies, which end with the same plant damage states (although the accident sequences are not necessarily mutually exclusive)

$$
F_{nss}(t) = \sum_{e=1}^{E} \sum_{f=1}^{F} F_{ASef}(t) \tag{5}
$$

where $F_{nss}(t)$ is the frequency of accident state (the plant damage state frequency) at time t, $F_{ASef}(t)$ the frequency of fth accident sequence in eth event tree at time t, E the number of event trees (equals to number of initiating events), and F is the number of accident states in event tree.

The accident frequency combines together the plant damage state frequencies

$$
F_n(t) = \sum_{d=1}^{D} F_{nssd}(t) \tag{6}
$$

where $F_n(t)$ is the accident frequency at time t (e.g. core damage frequency (CDF) at time t), $F_{nssd}(t)$ the frequency of dth accident state at time t, and D is the number of accident states.

2.2. Simulated annealing

Simulated annealing is a method suitable for solving optimization problems of a large scale special ones where a desired global extremum is hidden among many local extrema [18]. The idea of the method is an analogy with thermodynamics, specifically with the way that liquids freeze and crystallize, or metals cool and anneal. For slowly

cooled systems, nature is able to find the minimum energy state. If a liquid metal is cooled quickly it does not reach this state, but rather ends up in a polycrystalline or amorphous state having higher energy. So slow cooling is essential for ensuring that a low energy state is achieved.

Metropolis et al. [26] incorporated this kind of principles into numerical calculations. Offered a succession of options, a simulated system is assumed to change its configuration from energy $E1$ to energy $E2$ with probability p (where T is the temperature and k is the constant, which relates temperature with energy):

$$p = \begin{cases} e^{(-(E2-E1)/kT)} & \text{if } E1 < E2 \\ 1 & \text{if } E1 \geq E2 \end{cases} \quad (7)$$

The system goes downhill with probability of 1 for $E2 < E1$ (and for $E2 = E1$) and sometimes the system goes uphill if $E2 > E1$, but the lower the temperature, the less likely is any significant uphill excursion.

Metropolis algorithm provide four elements:

1. Description of possible system configurations.
2. A generator of random changes in the configuration; these changes are the options presented to the system.
3. An objective function: E (analog to energy), whose minimization is the goal of the procedure.
4. A control parameter T (analog to temperature) and an annealing schedule which tells how it is lowered from high to low values, e.g. after how many random changes in configuration is each downward step in T taken, and how large is that step.

The four elements of Metropolis algorithm applied to models of probabilistic safety assessment include:

1. Description of possible system configurations includes the mathematical model of probabilistic safety assessment. Possible system configurations, among which the optimum is searched, are calculations of CDF (CDF, as an example of accident frequency, F_n) at the plant level or fault tree top event probability (Q_{GD}) at the system level.
2. A generator of random changes in the configuration changes the changing vector of test placement times (T_{pBj}) of the components.
3. An objective function is CDF at the plant level (or Q_{GD} at the system level), whose minimization is the goal of the procedure.
4. A control parameter te (analog to temperature) and an annealing schedule (te = te × tefact; tefact < 1) tell how temperature is lowered from high to low values.

3. Optimization algorithm

The optimization objective is minimization of estimated risk through the minimization of the accident frequency (F_n), e.g. CDF at the plant level or minimization of the fault tree top event probability (Q_{GD}) of the respective system at the system level.

The objective function of optimization is represented by the mean value of obtained time dependent function over determined time interval $(0,\ldots,T)$. Possible system configurations, among which the optimum is searched, are calculations of CDF or Q_{GD} with respect to changing vector of test placement times (T_{pBj}) of the components. Fig. 1 shows the flowchart of the optimization algorithm.

3.1. System level

The fault tree time dependent top event probability $Q_{GD}(t, Q_{Bj}(t, T_{pBj}))$ represents the standpoint for evaluation of the system under investigation. The mean value (Q_{GD} mean) over determined time interval $(0,\ldots,T)$ represents the standpoint for the objective function

$$Q_{GDmean} = \frac{1}{T}\int_0^T Q_{GD}(t, Q_{Bj}(t, T_{pBj}))dt \quad (8)$$

where Q_{GDmean} is the mean fault tree top event probability, T the considered time interval, $Q_{GD}(t, Q_{Bj}(t, T_{pBj}))$ the time dependent top event probability as a function of time t and time dependent component failure probabilities $Q_{Bj}(t, T_{pBj})$, and $Q_{Bj}(t, T_{pBj})$ is the vector of time dependent component failure probabilities, which are function of test placement times (T_{pBj}) for all considered components Bj ($j \in \{1,\ldots,N\}$).

In the case of discrete time scale Eq. (8) is transformed into Eq. (9), where the time intervals ΔT between the time points, at which calculation is made, should suit the smallest reasonable time for activities in the plant to be modeled (ΔT is selected as 1 h):

$$Q_{GDmean} = \frac{1}{T}\sum_{t=1}^{t=T} Q_{GD}(t, Q_{Bj}(t, T_{pBj}) \quad (9)$$

The algorithm (Fig. 1) starts with reading input data for components, which are modeled as basic events in fault trees. Input data include parameters for every component (basic event), such as test intervals, failure rates and test duration times.

After initialization of defined variables, the two main loop portions repeat. First one represents calculations of system configurations at decreasing temperatures. Second one represents calculations performed at each temperature. If no better solution is found at certain temperature, the evaluation ends.

The central part starts with random selection of basic event index, which identify the component (basic event) and its appropriate random test placement time. The random selection of the test placement times is weighted with the length of test intervals. The test placement time of certain component cannot exceed its test interval. The weighted selection prevents that the test placement times for components with small test interval would be iterating too many times.

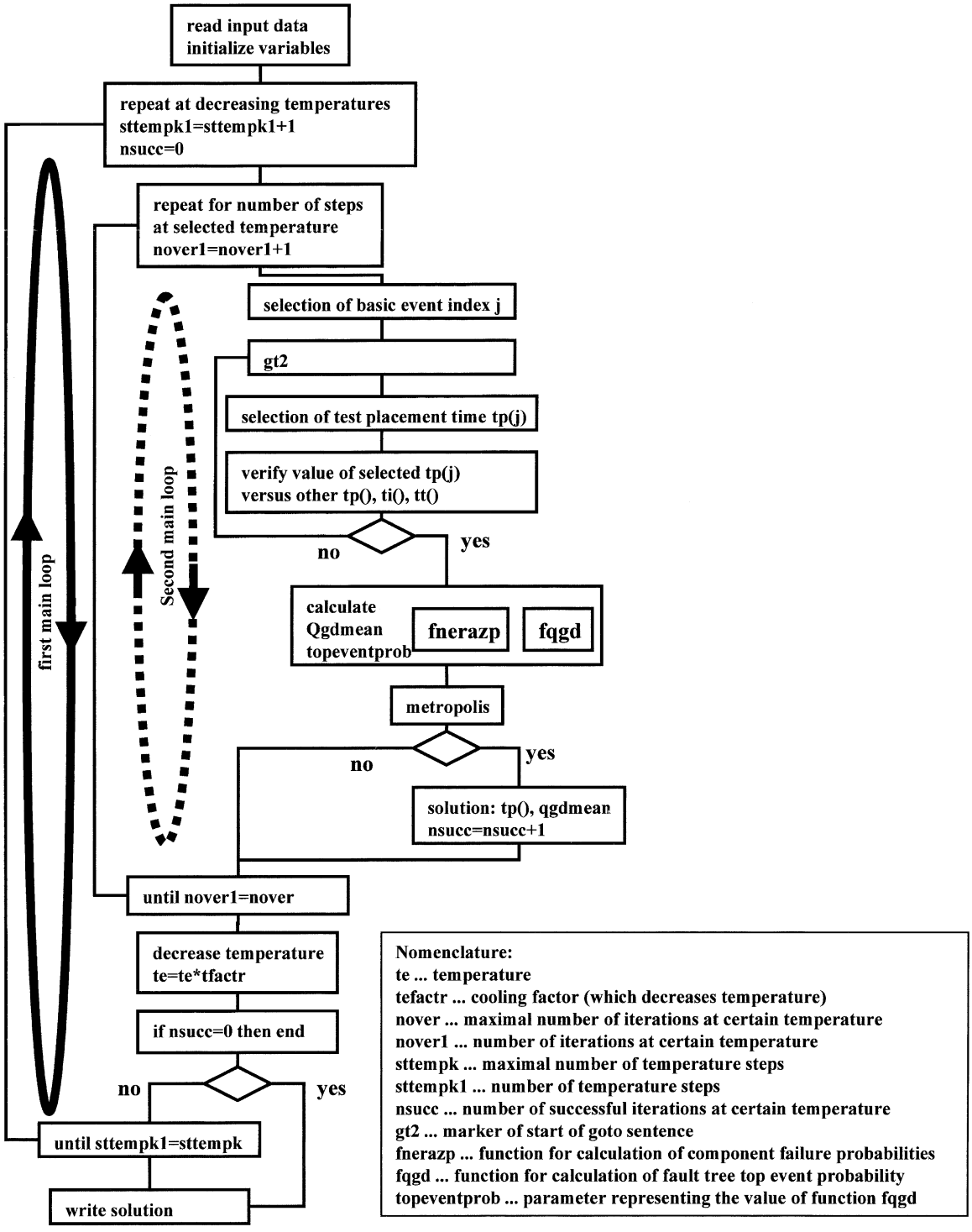


Fig. 1. The flowchart of the optimization algorithm.

The next step of the algorithm prevents undesired states (e.g. simultaneous tests) by comparison of the test placement times with the defined constraints.

The time dependent component failure probabilities $Q_{Bj}(t, T_{pBj})$ are calculated in each time point with the function fnerazp (Fig. 1). The function fqgd in all time points calculates the time dependent values of the fault tree top event probability based on the input minimal cut sets obtained from fault tree analysis.

The resulted mean value represents the objective function, which is the fault tree top event probability. The mean value is led to metropolis, which based on comparison with previous values determines the temporal solution. The temporal solution becomes the final solution after all

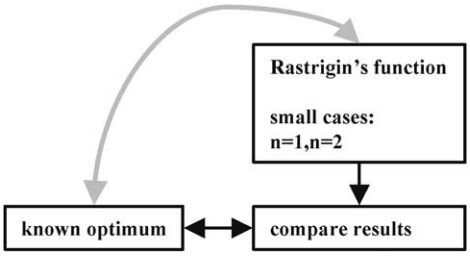


Fig. 2. Minimizing the Rastrigin's function.

evaluations at defined number of decreasing temperatures and after defined number of evaluations at defined temperature or when no improvements in the objective function is calculated with decrease of the temperature.

3.2. Plant level

The optimization algorithm at the plant level is analog to the algorithm at the system level. Objective function of the fault tree top event probability is in this case replaced by the objective function of accident frequency (F_n), e.g. CDF:

$$\mathrm{CDF}_{\mathrm{mean}} = F\mathrm{nmean} = \frac{1}{T}\sum_{t=1}^{t=T} F_n(t, Q_{Bj}(t, T_{pBj})) \qquad (10)$$

Optimization algorithm is focused to find its minimum (Fig. 1).

4. Performance testing and verification of the algorithm

After developing the algorithm for minimization of risk, which is based on finding the optimal test placement times, and which is described in Section 3.1, the same algorithm was examined for minimization of the generalized Rastrigin's function [27–29]. Generalized Rastrigin's function is highly multimodal function with many local extrema and one known global one:

$$f(xi) = 10n + \sum_{i=1}^{n} [(xi - xc)^2 - 10\cos(2\pi(xi - xc))] \qquad (11)$$

Algorithm found the global optimum, which is: $f(xi) = 0$ at $xi = xc$ (xc was selected as $xc = 100$). Fig. 2 shows that the solution was known and the results of the problem case evaluated by the algorithm were compared to known optimum for verification. Fig. 3 shows the results of algorithm runs for minimization of Rastrigin's function to show the number of successful iterations on the way to the global optimum.

After the performance tests with Rastrigin's function, the simulated annealing algorithm for minimization of risk was verified by comparison of its results to three other algorithms:

(a) algorithm for examination of all possible combinations,
(b) algorithm based on random trials and
(c) algorithm based on component by component determination of minimum.

Fig. 4 shows, that the results were compared to known optimum. The examination of all possible combinations of test placement times gave the known optimum i.e. minimum of risk. This was possible only for small cases with small number of components and with their small test intervals.

Results of evaluations performed at small number of components (number of components: jj = 4) and at small test intervals were the same for all four algorithms (time interval: ttt = 10). This increased the confidence in correctness of the simulated annealing algorithm for minimization of risk for the evaluation of real cases, which are bigger in that extent that all possible configurations cannot be

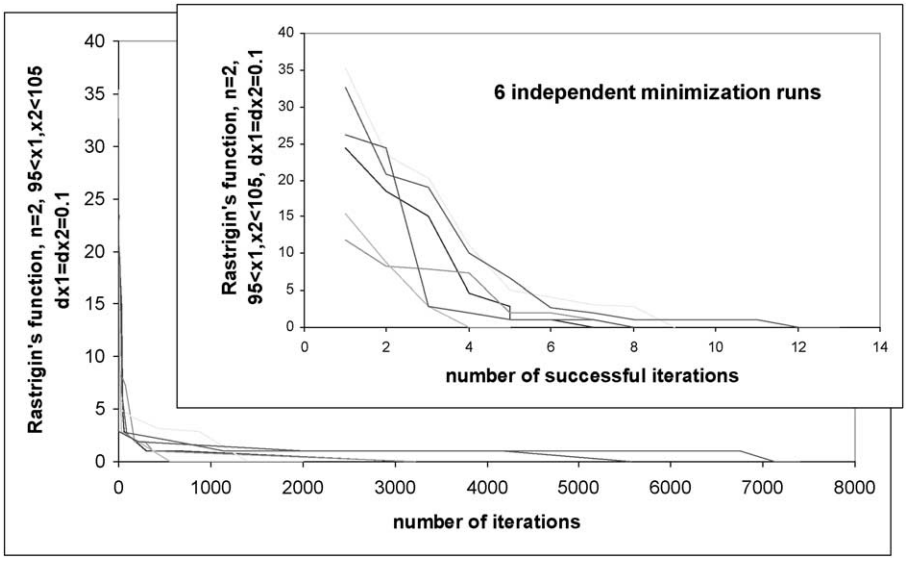


Fig. 3. Results of minimization of Rastrigin's function.

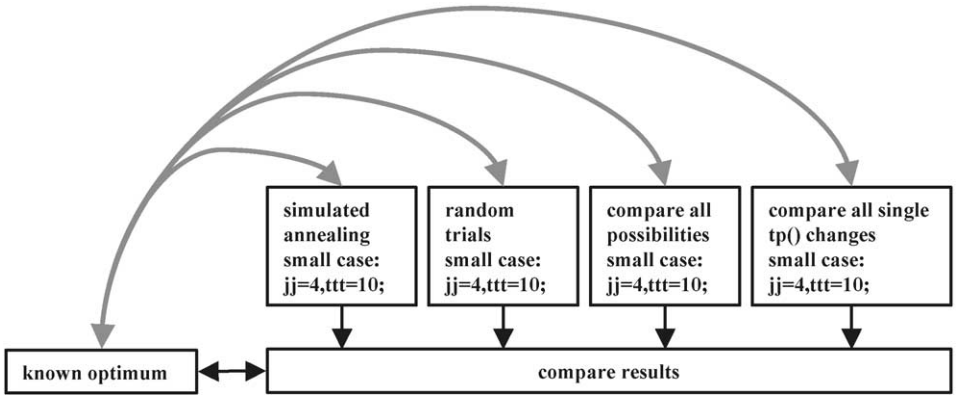


Fig. 4. Verification of simulated annealing algorithm.

evaluated and verified similarly as it is the case for small examples.

5. Results

The simulated annealing algorithm for minimization of risk was used for evaluation of several examples. The first one represents a system consisting of 10 components in series. The second one represents the high-pressure injection system shown by Harunuzzaman [17] from NT96, which consists of seven valves and three pumps, which provide water to two injection paths. Benefit from its application is in avoiding of schedules of equipment outages that may lead to the higher risk and thus in the overall risk reduction with identification of schedules of equipment outages that may lead to the lower risk.

5.1. Evaluation results of a system consisting of 10 components in series

Evaluation results of the system consisting of 10 components connected in series show that mean fault tree top event probability may be reduced for approximately 6%. Fig. 5 shows the convergence of the objective function for such a system. Fig. 6 shows the time dependent top event probability of the optimal schedule, i.e. schedule with determined optimal test placement times for all 10 considered components. Table 1 shows the data, which was used for probabilistic modeling of components as the input for optimization.

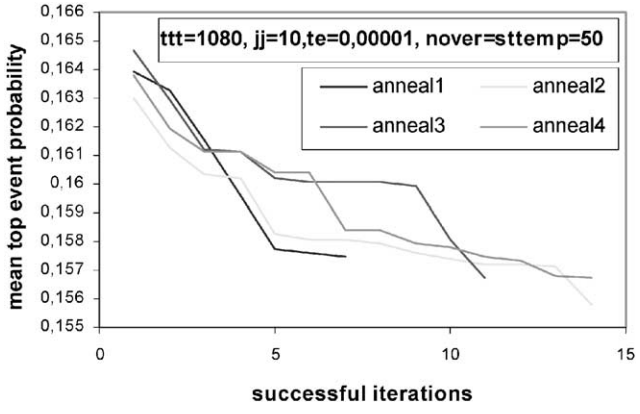


Fig. 5. Convergence of the objective function for system consisting of 10 components in series (four independent minimization runs: anneal{1,2,3,4}).

5.2. Evaluation results of the example system shown by Harunuzzaman [17] from NT96

Evaluation results of the high-pressure injection system [17], which consists of seven valves and three pumps, which provide water to two injection paths, show that mean top event probability may be reduced for approximately 5%. Fig. 7 shows the convergence of the mean top event probability as the objective function for the system. Fig. 8 shows the time dependent top event probability of the optimal schedule, i.e. schedule with determined optimal test placement times for all 10 considered components. Table 2 shows the data, which was used for probabilistic modeling of components as the input for optimization.

6. Discussion

Experience shows that the testing and maintenance activities of safety equipment in nuclear power plants represent an important potential for risk and cost reduction. Time scheduling of safety equipment outages due to the testing and maintenance was selected to reduce risk.

Models and results of probabilistic safety assessment serve as a standpoint, because they are a comprehensive source of safety information, although the classical models need updating in a sense that they are able to model the time requirements. Alternatively, the contributions of testing and maintenance may be extracted from the entire probabilistic safety assessment model and be evaluated separately. Such alternative results in larger relative differences in evaluated risk measures between equipment outage arrangements, but the results are equally applicable and evaluation is faster.

Simulated annealing as a powerful optimization method was selected to find plant configuration resulting in minimal risk, because it is an extremely useful method for solving

Table 1
Data for probabilistic modeling of the system consisting of 10 components in series (components named: Be1,Be2,…,Be10 are modeled in the respective basic events; T_p^*—initial T_p, T_p^{**}—optimal T_p)

Component name	Failure rate (h^{-1})	T_i (h)	T_p^* (h)	T_p^{**} (h)	T_t (h)	Probility/demand
Be1	1.00×10^{-4}	100	1	26	1	1.00×10^{-3}
Be2	1.00×10^{-4}	100	2	79	1	1.00×10^{-3}
Be3	1.00×10^{-4}	100	3	40	1	1.00×10^{-3}
Be4	1.00×10^{-4}	100	4	88	1	1.00×10^{-3}
Be5	1.00×10^{-4}	100	5	87	1	1.00×10^{-3}
Be6	1.00×10^{-4}	200	7	132	2	1.00×10^{-3}
Be7	1.00×10^{-4}	200	8	157	1	1.00×10^{-3}
Be8	1.00×10^{-4}	200	10	107	2	1.00×10^{-3}
Be9	1.00×10^{-4}	200	11	128	1	1.00×10^{-3}
Be10	1.00×10^{-4}	200	13	135	2	1.00×10^{-3}

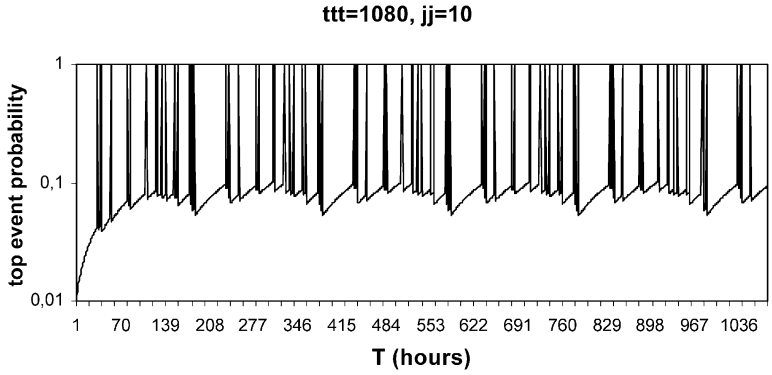


Fig. 6. Time dependent top event probability at optimal schedule of the system consisting of 10 components in series.

optimization problems of large scale with desired global extremum hidden among many local extrema.

Performance tests successfully served for verification and assessment of the algorithm. They were performed on Rastrigin's multimodal function with a large number of local extrema for small and large cases. In addition, the algorithm for minimization of risk was verified by comparison of its results to three other algorithms of which the examination of all possible combinations is possible only for very small cases because of fast increasing of combinations with extent of the case.

When running performance tests in larger, multidimensional cases (e.g. $n = 100$) of Rastrigin's function, the algorithm sometimes ended in the local optimum very close to the global one, because the selected number of iterations was smaller than necessary to get the global optimum. The number of iterations was not increased and the algorithm was not improved (e.g. with the root increase of cooling rate [30]) because of two reasons. Firstly, the important goal was to get quick result. Secondly, the nature of the primary objective function is such that many local extrema minimally differ one from another and from the global one. Because of large uncertainties of the results of probabilistic safety assessment it was expected that it does not make any sense to prolong the evaluations to improve resulted risk measure on its second or third decimal.

Results of several examples showed that it is possible to reduce risk by application of the developed algorithm, which deals with optimization of schedule of safety equipment outages. The results confirmed that considering large uncertainties in probabilistic safety assessment may lead to the conclusion, that the most important result of the method is not a selection of the most suitable schedule among those, which results in similarly low risk. But, it may be a prevention of such schedules, which result in high risk. Such

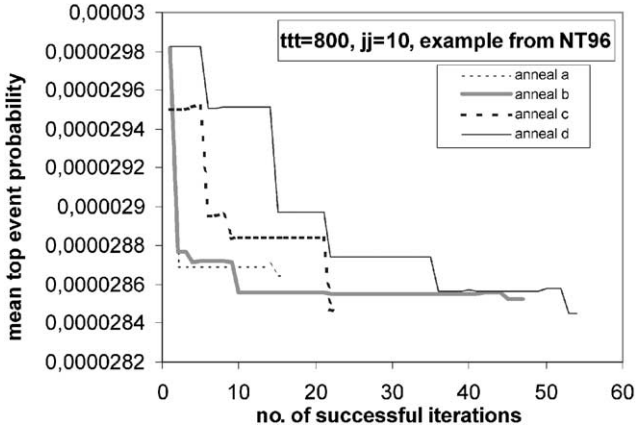


Fig. 7. Convergence of the objective function (example system from NT96; four independent minimization runs: anneal{a,b,c,d}).

Table 2
Data for probabilistic modeling of the system (example system from NT96) (components named: Be1-V1,Be2-V2,…,Be10-P3, are modeled in the respective basic events; together there are seven valves and three pumps; T_p^*—initial T_p, T_p^{**}—optimal T_p)

Component name	Failure rate (h^{-1})	T_i (h)	T_p^* (h)	T_p^{**} (h)	T_t (h)	Probability/demand
Be1-V1	5.83×10^{-6}	200	1	44	1	1.82×10^{-3}
Be2-V2	5.83×10^{-6}	200	2	17	1	1.82×10^{-3}
Be3-V3	5.83×10^{-6}	200	3	108	1	1.82×10^{-3}
Be4-V4	5.83×10^{-6}	200	4	72	1	1.82×10^{-3}
Be5-V5	5.83×10^{-6}	200	5	176	1	1.82×10^{-3}
Be6-V6	5.83×10^{-6}	200	7	153	1	1.82×10^{-3}
Be7-V7	5.83×10^{-6}	200	8	149	1	1.82×10^{-3}
Be8-P1	3.89×10^{-6}	400	10	247	2	5.30×10^{-4}
Be9-P2	3.89×10^{-6}	400	12	198	2	5.30×10^{-4}
Be10-P3	3.89×10^{-6}	400	14	107	2	5.30×10^{-4}

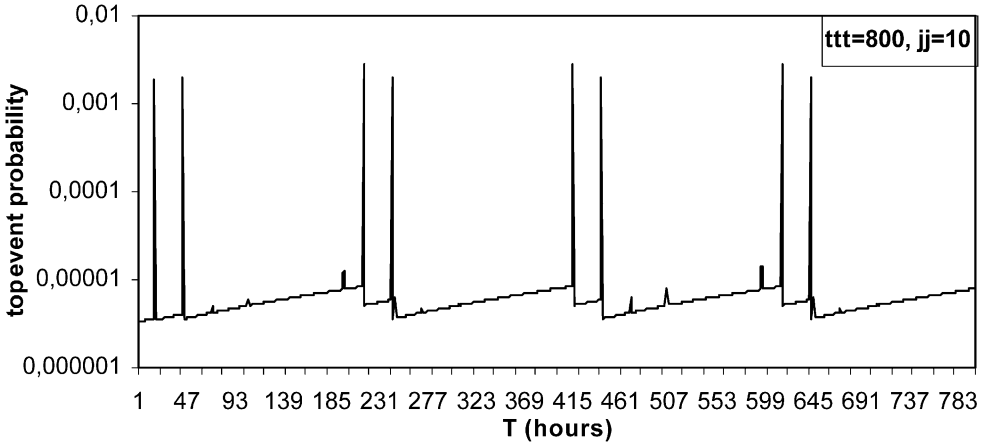


Fig. 8. Time dependent top event probability at optimal schedule (example system from NT96).

finding increases the importance of evaluation speed versus the requirement of getting always the global optimum no matter if it is only slightly better that certain local one. So the number of evaluation iterations may be reduced from their number, which is required to get the global optimum, to get fast evaluation results.

The weak point of optimization is the selection and application of time dependent probabilistic models because they have been not developed yet in extend to model all contributions (positive and negative) of testing and maintenance, specially including time dependency.

Nevertheless, the approach has a perspective in future, specially, in direction of integrating power and shutdown probabilistic safety assessment, which is theoretically possible with the use of dynamic fault tree [20]. Integration of modes of operation would allow possibility of evaluating the risk contributions of testing and maintenance activities between the modes of plant operation, e.g. towards the on-line maintenance.

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References

[1] Čepin M, Leskovar M, Prošek A, Mavko B. Development of new methods and models for improvement of nuclear safety (in Slovenian), Naravoslovje in tehnika, DMRS, 2000. p. 239–49.
[2] PRA Procedures Guide. NUREG/CR-2300, US NRC; 1982.
[3] Čepin M, Gomez Cobo A, Martorell S, Samanta P. Methods for testing and maintenance of safety related equipment: examples from an IAEA Research Project. Proc ESREL99: Safety Reliab 1999;1:247–51.
[4] Aven T, Dekker R. A usefull framework for optimal replacement models. Reliab Engng Syst Safety 1997;58:61–7.
[5] Schabe H. A new approach to optimal replacement times for complex systems. Microelectron Reliab 1995;35(8):1125–30.
[6] Chelby A, Ait-Kadi D. An optimal inspection strategy for randomly failing equipment. Reliab Engng Syst Safety 1999;63:127–31.
[7] Martorell S, Carlos S, Sanchez A, Serradell V. Constrained optimization of test intervals using a steady-state genetic algorithm. Reliab Engng Syst Safety 2000;67:215–32.
[8] Čepin M, Mavko B. Probabilistic safety assessment improves surveillance requirements in technical specifications. Reliab Engng Syst Safety 1997;56:69–77.
[9] Vaurio JK. Optimization of test and maintenance intervals based on risk and cost. Reliab Engng Syst Safety 1995;49:23–36.
[10] Yang JE, Sung TY, Yin Y. Optimization of the surveillance test interval of the safety systems at the plant level. Nucl Technol 2000;132:352–65.
[11] Uryasev S, Vallerga H. Optimization of test strategies: a general approach. Reliab Engng Syst Safety 1993;41:155–65.
[12] Hokstad P, Flotten P, Holmstrom S, McKenna F, Onshus T.

A reliability model for optimisation of test schemes for fire and gas detectors. Reliab Engng Syst Safety 1995;47:15–25.
[13] Čepin M. Sequential versus staggered testing towards dynamic PSA. Proceedings of the Second Regional Meeting on Nuclear Energy in Central Europe, NSS and ENS, Portorož; September 11–14, 1995. p. 184–9.
[14] Fabjan L, Jordan-Cizelj R, Čepin M. Assessment of Krško NPP—technical specification changes related to the surveillance requirements, IJS-DP-7737; 1998.
[15] Kang DI, Kim SD, Jang SC, Kim KY. Evaluation of risk on the preventive maintenance of ulchin units 3 and 4 during power operation, PSAM5; 2000.
[16] Hook T, Henneke D, Morgan T. Insights from a risk-informed standardized technical specifications pilot project at San Onofre Nuclear Generating Station, PSAM5; 2000.
[17] Harunuzzaman M, Aldemir T. Optimization of standby safety system maintenance schedules in nuclear power plants. Nucl Technol 1996;113(March):354–67.
[18] Martorell S, Sanchez A, Carlos S, Seradell V. Simultaneous and multi-criteria optimization of TS requirements and maintenance at NPPs. Ann Nucl Energy 2002;29(2):147–68.
[19] Press WH, Teukolsky AS, Vetterling WT, Flannery BP. Numerical recipes in Fortran. Cambridge: Cambridge University Press, 1992.
[20] Čepin M, Mavko B. A dynamic fault tree. Reliab Engng Syst Safety 2002;75(1):83–91.
[21] Caruso MA, Cheok MC, Cunningham MA, Holahan GM, King TL, Parry GW, Ramey-Smith AM, Rubin MP, Thadani AC. An approach for using risk assessment in risk-informed decisions on plant-specific changes to the licensing basis. Reliab Engng Syst Safety 1999;63:231–42.
[22] Čepin M, Mavko B. Fault tree developed by an object-based method improves requirements specification for safety-related systems. Reliab Engng Syst Safety 1999;63:111–25.
[23] Roberts NH, Vesely WE, Haasl DF, Goldberg FF. Fault tree handbook, NUREG-0492, US NRC, Washington; 1981.
[24] Čepin M. Estimation of event frequencies on the various system and plant levels. EUROCOURSE, PSARID, Garching; 2001.
[25] Lofgren EV, Cramond WR. Probabilistic risk assessment course documentation, vols. 1–7, NUREG/CR-4350; 1985.
[26] Metropolis N, Rosenbluth A, Rosenbluth M, Teller A, Teller E. Equations of state calculations by fast computing machines. J Chem Phys 1953;21:1087–92.
[27] Hart WE. Adaptive global optimization with local search. Doctoral Thesis, University of California; 1994.
[28] Rastrigin LA. Systems of Extremal Control. Nauka; 1974.
[29] Mühlenbein H, Schomisch M, Born J. The parallel genetic algorithm as function optimizer. Proc Fourth Int Conf Genetic Algorithms 1991:271–8.
[30] Zhu J, Bilbro G, Chow MY. Phase balancing using simulated annealing. IEEE Trans Power Syst 1999;14(4):a.